TCP / UDP Echo

Sample input (client):

Enter message: Hello server

output client:

connected to server

received replied Hello Server.

output (Server):

Server is listening

client connect

Received: Hello server.

Echo sent successfully



Exp: 12: End to End communication at the Transport layer.

a) Implement Echo client server using TCP UDP sockets.

Aim: To implement an echo client, server application using TCP and UDP sockets and demonstrate end to end communication at the transport layer.

Algorithm:

1) Choose a programming language such as a python and import the socket module

2) Create a server socket using TCP (SOCK STREAM) or UDP. (SOCK DGRAM)

3) Assign an IP address and port number the servers

4) for UDP bind the socket and wait for incoming ad datagrams

5) Create a client socket and connect to the server. for TCP or Send a datagram for UDP

6) Read a message entered by the client ad to the server

7) Make the server return the received message to client.

Sample input (client)

Enter message : Hi

sample output (server)

client : Hi

server : Hello :

Sample output (client):

Server : Hello

b) implement a chat program using Socket Programming

AIM : To implement a client server chat application using socket programming for exchanging message btw a computer

Algorithm :

1) import the Socket module in the selected programming language.

2) create a server socket and bind it to suitable IP address

3) Allow both the client and server to enter messages

4) Transmit the message using the socket's send and receive operation.

5) Display the received message on the respective terminals.

6) continue communication until either side enter an exit command.

Result :

Thus chat application was implemented successfully using socket programming. The client and server exchanged messages through a